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Activity

1. Students are asked to identify the unlabelled slides of tissues and to classify them. Similar exercise can also be accomplished by projecting unlabelled histological images on a screen. They can identify the slides of different tissues through microscope
2. The preparation of smear of stratified squamous epithelia from the inner lining of cheek allows the students to make their own slides using biological stain. They will have the experience of examining their cheek cells.

Summary

The body cells combine to form four different types of tissues; **Epithelial, Connective, Muscle and Nervous tissues.** Though the cells of these tissues share certain features in common, by no means they are identical. They belong together because they have basic fundamental resemblances. The important concept to carry away with you is that tissues, despite their unique abilities, cooperate to keep the body safe, healthy, viable and whole.

Glossary

Acinus – Cells arranged into a circular secretory unit

Adipocyte – Large cell (up to 200 microns) with only a thin film of cytoplasm due to the presence of a large fat droplet.

Adipose tissue – A group of adipocytes.

Bone – Specialized connective tissue with a mineralized (hydroxyapatite) matrix.

Collagen – A triple helix protein which allows for great tensile strength.

Goblet cell – special mucus secreting columnar epithelial cell located in the respiratory tract and intestine.

Lacunae – A cavity or depression especially in the bone

Mast cells – Cells filled with basophilic granules found in numbers in connective tissue and releases histamine and other substances during inflammatory and allergic reactions.

Macrophages – Immune cells derived from monocytes; engaged in phagocytosis of microbes and debris.

Evaluation

1. The main function of the cuboidal epithelium is
   a. Protection b. Secretion
   c. Absorption d. Both (b) and (c)
2. The ciliated epithelium lines the
   a. Skin b. Digestive tract
   c. Gall bladder d. Trachea
3. What type of fibres are found in connective tissue matrix?
   a. Collagen b. Areolar
   c. Cartilage d. Tubular